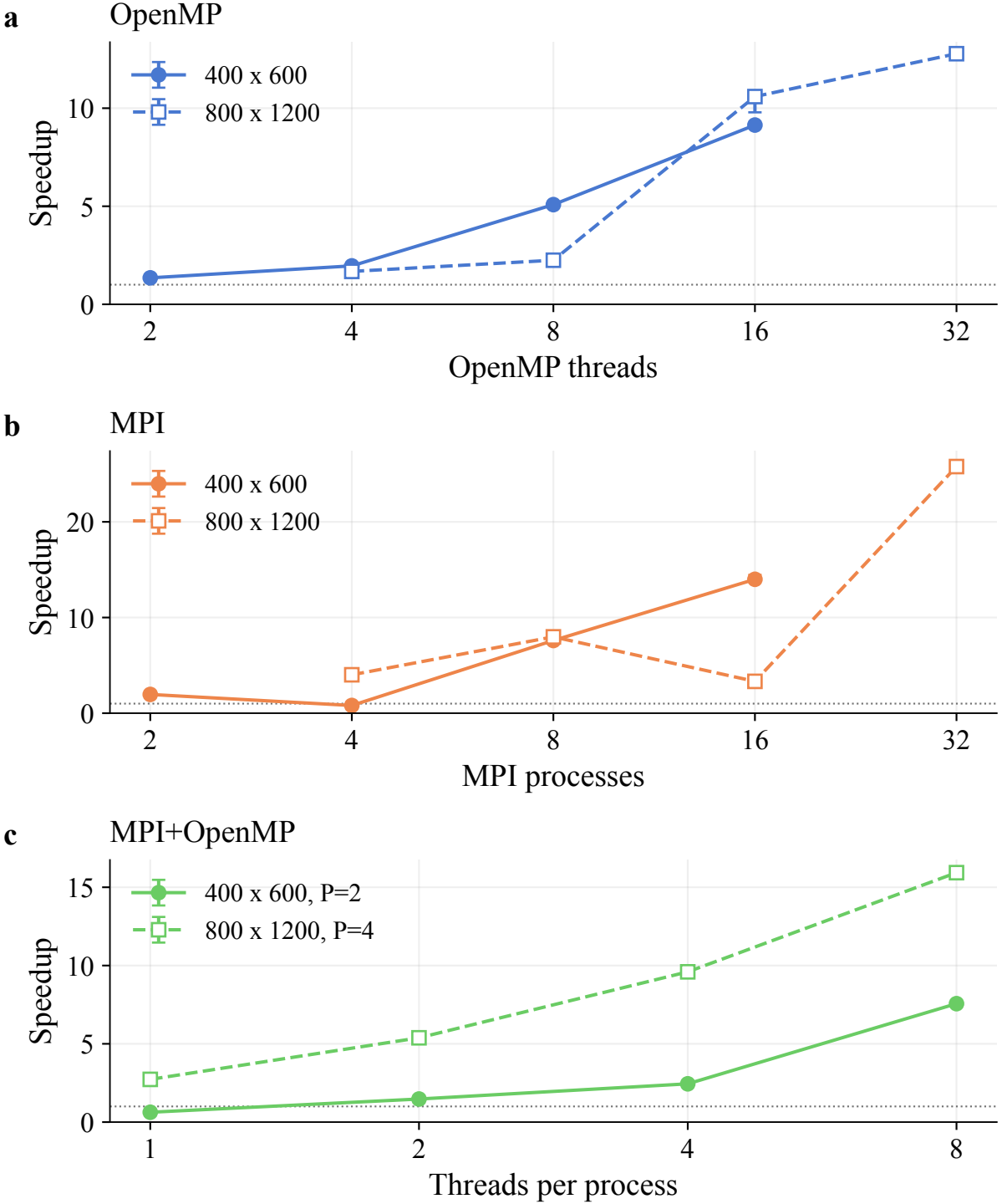
a
OpenMP
400 x 600
800 x 1200
Speedup
0
5
10
2
4
8
16
32
OpenMP threads
b
MPI
400 x 600
800 x 1200
Speedup
0
10
20
2
4
8
16
32
MPI processes
c
MPI+OpenMP
400 x 600, P=2
800 x 1200, P=4
Speedup
0
5
10
15
1
2
4
8
Threads per process